

Simple 5W Stereo Amplifier For Noise-Free Music

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This simple stereo audio amplifier is based on a 12-pin TA8207K IC, which is inexpensive and easily available. In addition, it requires only a few resistors and electrolytic and ceramic capacitors. You can get output power of about 4.6W for each (left and right) channel using a 12V DC supply.

Circuit and working

The author's prototype is shown in Fig. 1 and the circuit diagram is shown in Fig. 2.

As shown in Fig. 2, pins 3 and 12 of IC1 (TA8207K) are connected to +12V terminal of the battery. Pins 6 and 7 of the IC are used for left and right audio

inputs, respectively. A 100-kilo-ohm pot can be connected to each input of the amplifier for volume control (not shown in Fig. 2).

Loudspeakers LS1 and LS2 are connected to pins 2 and 10 of IC1 through electrolytic capacitors C8 and C9, respectively. Electrolytic capacitor C10 connected between +12V and ground acts as a noise filter in the supply voltage.

Construction and testing

You can assemble this amplifier on a 5cm × 6.5cm Veroboard, or on the PCB whose layout is shown in Fig. 3, with its components layout in Fig. 4.

IC TA8207K, which can be vertically inserted into the Veroboard or PCB, will need a heat-sink to prevent damage to the IC due to overheating that can occur during operation. A small piece of aluminium sheet can be used as the heat-sink.

You may solder two 2-pin terminals (not shown in Fig. 2) for the right and left speakers on the PCB or Veroboard. Solder a 3-pin connector on the board for input signals and a small 2-pin terminal connector for the power supply.

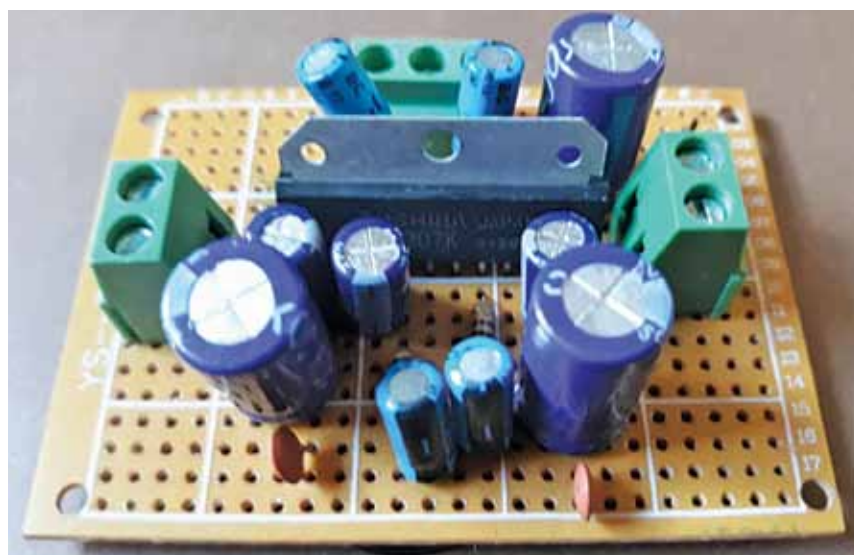


Fig. 1: Author's prototype

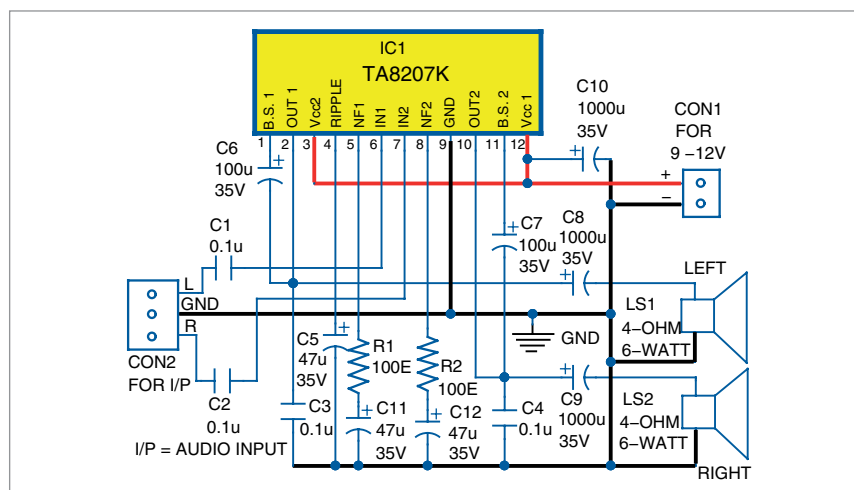


Fig. 2: Circuit diagram of the amplifier

PARTS LIST

Semiconductors:

IC1 - TA8207K stereo audio amplifier

Resistors (all 1/4-watt, ±5% carbon):

R1, R2 - 100-ohm

Capacitors:

C1-C4 - 0.1μF ceramic disk
C5, C11, C12 - 47μF, 35V electrolytic
C6, C7 - 100μF, 35V electrolytic
C8-C10 - 100μF, 35V electrolytic

Miscellaneous:

CON1 - 2-pin terminal
CON2 - 3-pin connector
LS1, LS2 - 4-ohm, 6-watt speaker
- Heat-sink for the IC
- Stereo jack (not shown in PCB)
- 2-pin terminals (2 numbers) for LS1 and LS2